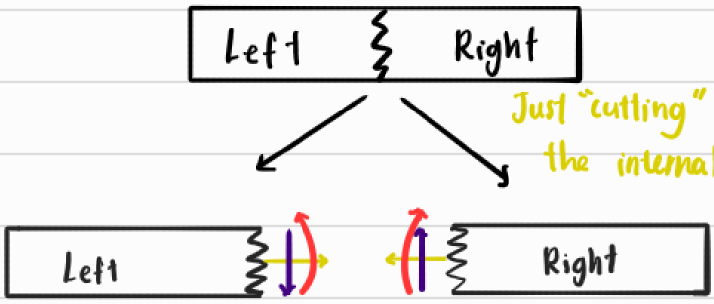
Sign Conventions of internal forces of beams



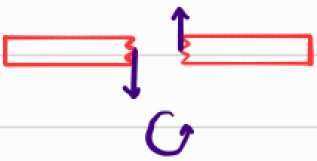
Just "cutting" it up to see the internal forces clearly
[It's not actually cut]

— Bending moment
— Shear force
— Axial Force

All these direc. are positive

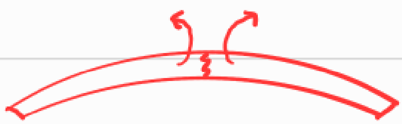
*Note :

① A +ve shear force means the same direc. as convention



This torque of a couple causes an anti-clockwise moment

② A +ve bending moment causes the beam to make a sad face

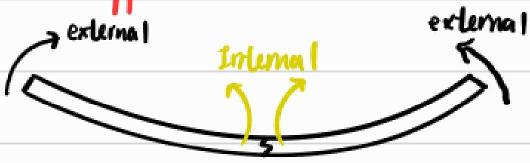


*However to take note that :

Internal forces are created to counteract external forces to maintain eq.

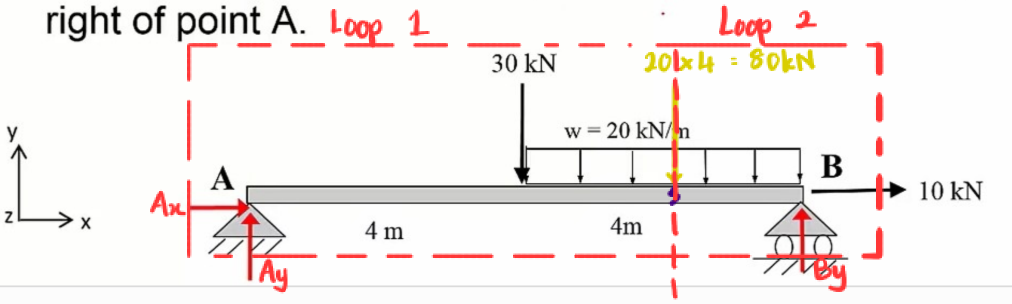
① This means a +ve shear force means that an external shear force in the opposite direc. have been applied

② A +ve bending moment means that a external bending moment have been applied in the opposite direction



Example 1 :

Determine the internal forces (bending, shear and axial) 6m right of point A.



Finding Rxn Forces at joints

$\Sigma M_A = 0 = -30(4) - [20 \times 4](6) + B_y(8)$

$B_y = 75\text{kN}$ #

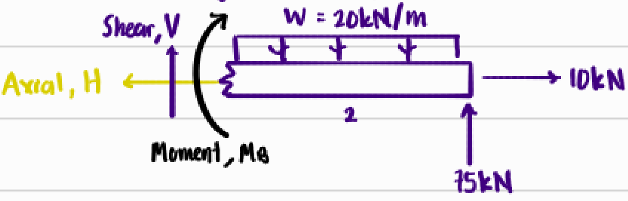
$\Sigma F_y = 0 = A_y - 30 - 80 + 75$

$A_y = 35\text{kN}$ #

$\Sigma F_x = 0 = A_x + 10\text{kN}$

$A_x = -10\text{kN}$ #

Looking at loop 2



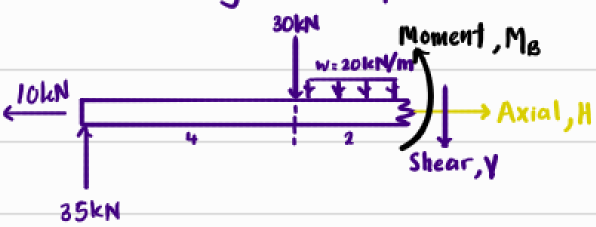
$\Sigma F_x = 0 = -H + 10 \Longrightarrow H = 10\text{kN}$ #

$\Sigma F_y = 0 = V + 75 - (20 \times 2) \Longrightarrow V = -35\text{kN}$

$\Sigma M_{cut} = 0 = -M_B - (20 \times 2)(1) + 75(2)$

$M_B = 110\text{kN m}$ #

Looking at loop 1



$\Sigma F_x = 0 = H - 10$

$H = 10\text{kN}$ #

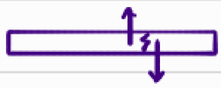
$\Sigma F_y = 0 = 35 - 30 - 20(2) - V$

$V = -35\text{kN}$ #

$\Sigma M_{cut} = 0 = M_B - 35(6) + 30(2) + (20 \times 2)(1)$

$M_B = 110\text{kN m}$ #

∴ Since our shear force is negative,



it causes clockwise rotation

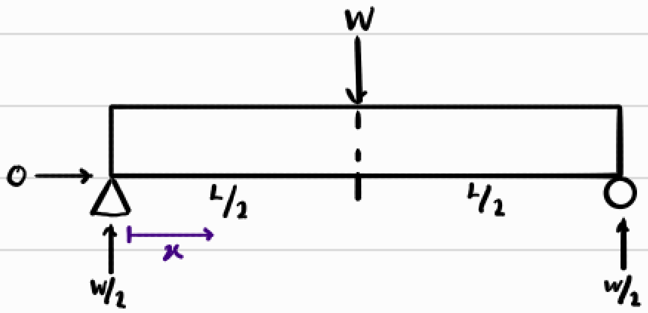
Since bending moments are positive, this means the beam is sagging

Since axial forces are +ve and away, this means the beam is in tension

Shear Force Diagrams, SFD

Since at every point of the beam might experience diff ammount of shear force, we can just generalise an eqn for it

Example 1:



* We included the rxn force at pin joint eventhough our interval didn't include $x = 0$ bcuz the interval describes where internal shear force exist, not external.
Without an external force, there wouldn't be internal forces

[B4 point load]

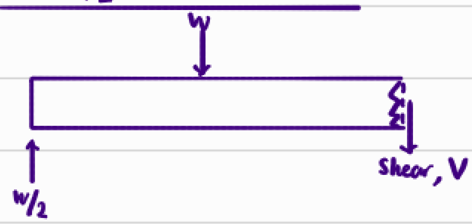
For $0 < x < L/2$

$\therefore \Sigma F_y = 0 = \frac{W}{2} - V$

$V = W/2$

[After point load]

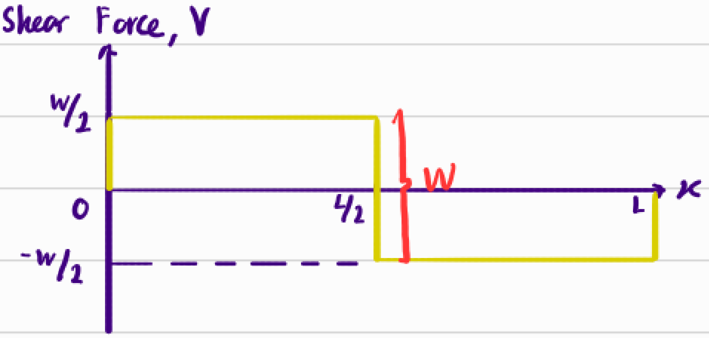
For $L/2 < x < L$



$\therefore \Sigma F_y = 0 = W/2 - W - V$

$V = -W/2$

We get the fllwing graph,



* Note:

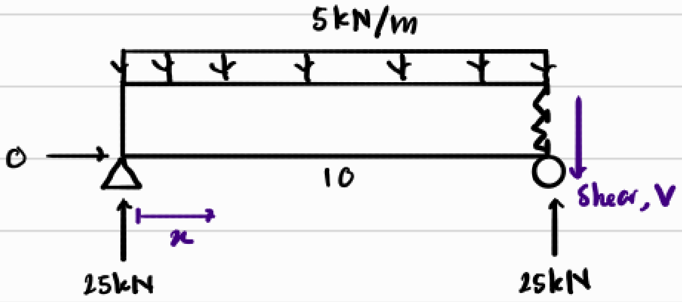
[Sometimes these are included for simplicity sake]

① $x = 0$ & $x = L$ is not included, bcuz that's not internal anymore, thus $V = 0$

② We did not include $x = L/2$ [Right b/w the point load] within our 2 intervals bcuz the shear force at that point is undefined.

As we can see, right before the point load, $V = W/2$, right after the point load, $V = -W/2$.
This must be true to maintain equilibrium. Thus, the V at $x = L/2$ has to "jump".

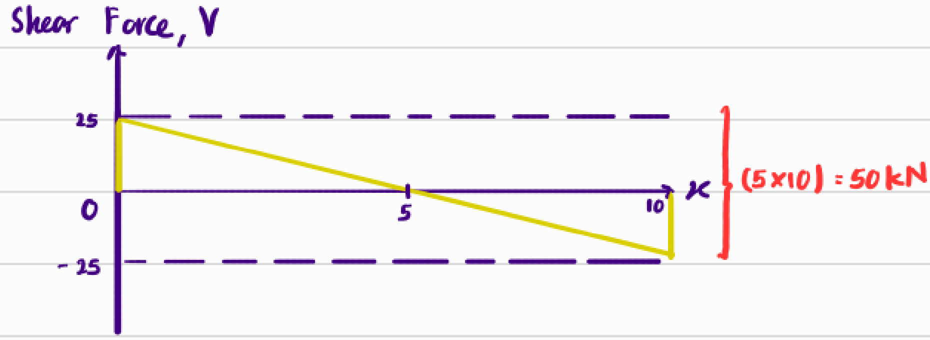
Example 2:



$0 < x < 10$

$\Sigma F_y = 0 = 25 - 5x - V$

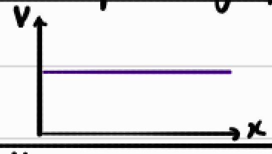
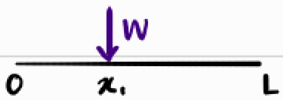
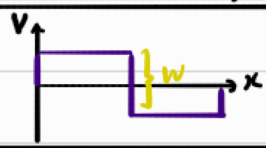
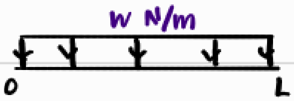
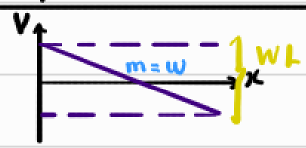
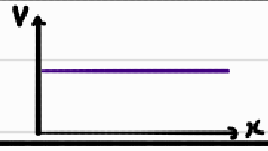
$V = -5x + 25$



* Notice that V doesn't jump. This is bcuz for a uniformly distributed load, as x ↑, the external force ↑ gradually.
As for a point load in the previous example, the external force increases instantly at the point load.

SFD shortcut

Since shear force is technically just the ΣF_y at every point, just in the opposite direc., we can plot the graph without calculation w̄ the table blw

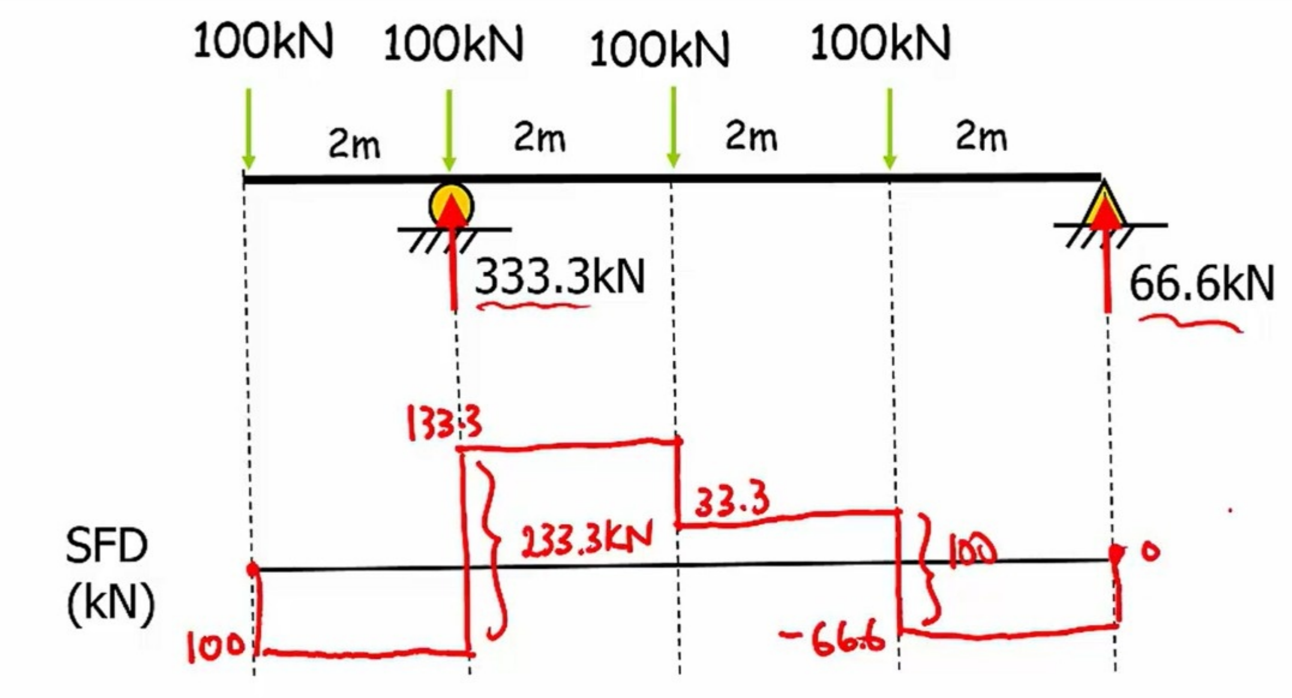
Type of load	Diagram of load	Shape of graph
No Load		
Point Load		
Uniformly distributed Load		
Applied Moment		

* When drawing the graph, if the external force is upwards, ΔV is +ve and vice versa. This is bcuz, we are always only just taking & increasing the length of the left cut piece. The convention of +ve shear force for the left cut piece is downwards.

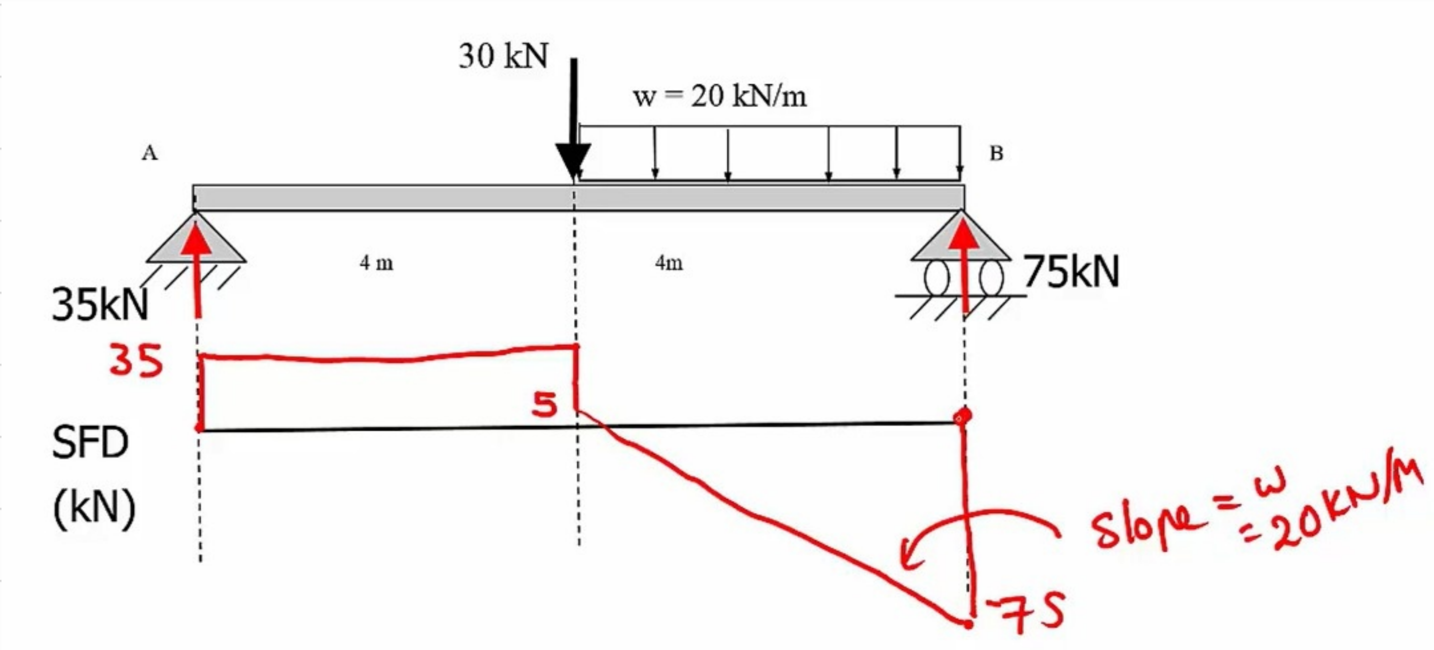
∴ When F_{ex} is upwards, V is downwards $\Longrightarrow$ +ve

F_{ex} is downwards, V is upwards $\Longrightarrow$ -ve

Example 1:



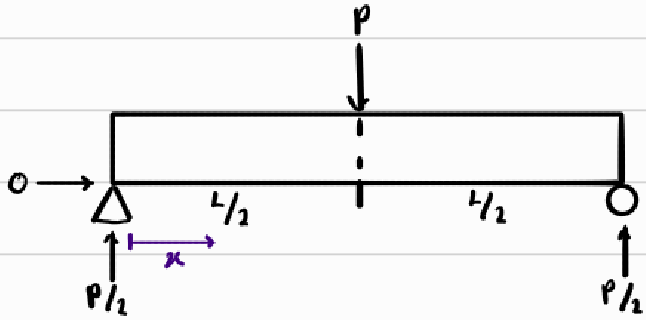
Example 2:



Bending Moment Diagram, BMD

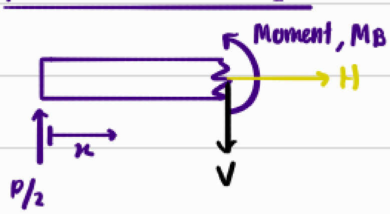
Since at every point of the beam might experience diff ammount of bending moment, we can just generalise an eqn for it

Example 1:



[B4 point load]

For $0 < x < L/2$

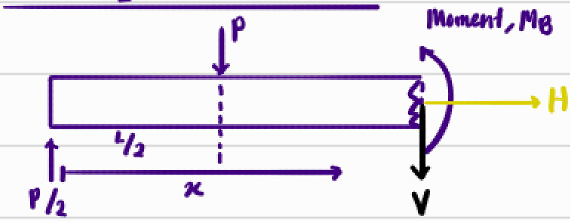


$\therefore \Sigma M_{cut} = 0 = M_B - \left(\frac{P}{2}\right)x$

$M_B = \left(\frac{P}{2}\right)x$

[After point load]

For $L/2 < x < L$



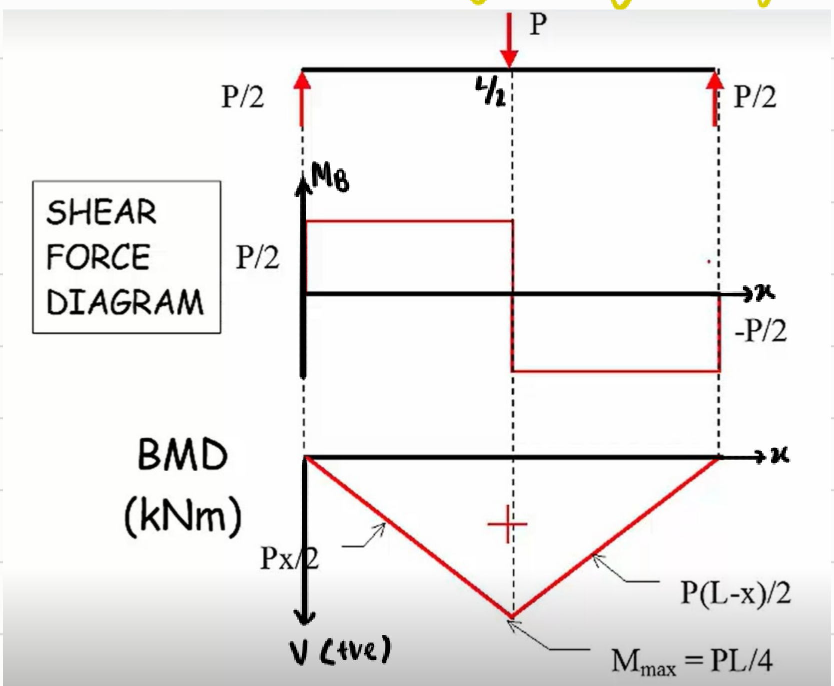
$\therefore \Sigma M_{cut} = 0 = M_B - \left(\frac{P}{2}\right)x + P\left[x - L/2\right]$

$M_B = \frac{P(L-x)}{2}$

Now for the sake of clarity, since we know that a +ve bending moment $\Longrightarrow$ sagging, and sagging goes downwards.

$\therefore$ For a +ve bending moment, we plot downwards as positive

[This makes it clearer by showing the general shape of the beam]

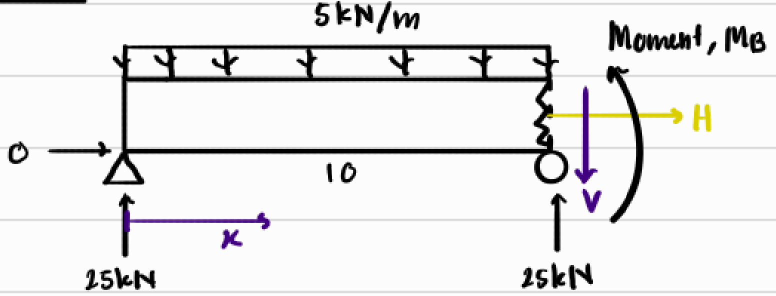


As u can see, we know that the beam will bent downwards

thus we plot downwards to show that

[Convention set by Monash]

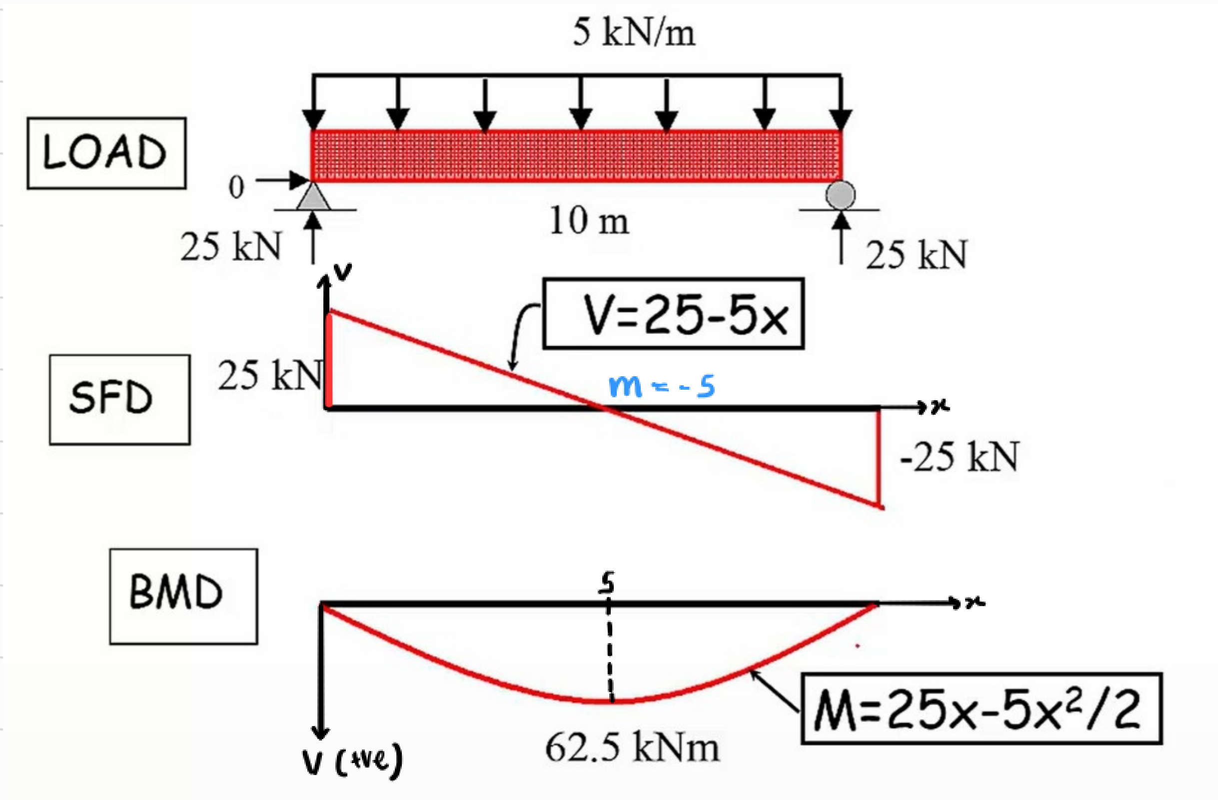
Example 2:



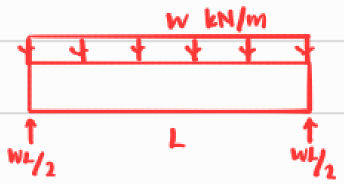
$0 < x < 10$

$$\Sigma M_{cut} = 0 = M_B - 25x + (5x)(x/2)$$

$$M_B = 25x - \left(\frac{5}{2}\right)x^2$$



Take Note:

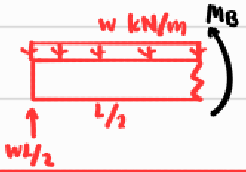


$$\Sigma M_{cut} = 0 = M_B + \left(w \times \frac{L}{2}\right)(L/4) - \left(\frac{wL}{2}\right)(L/2)$$

$$M_B = \frac{wL^2}{8} = M_{max}$$

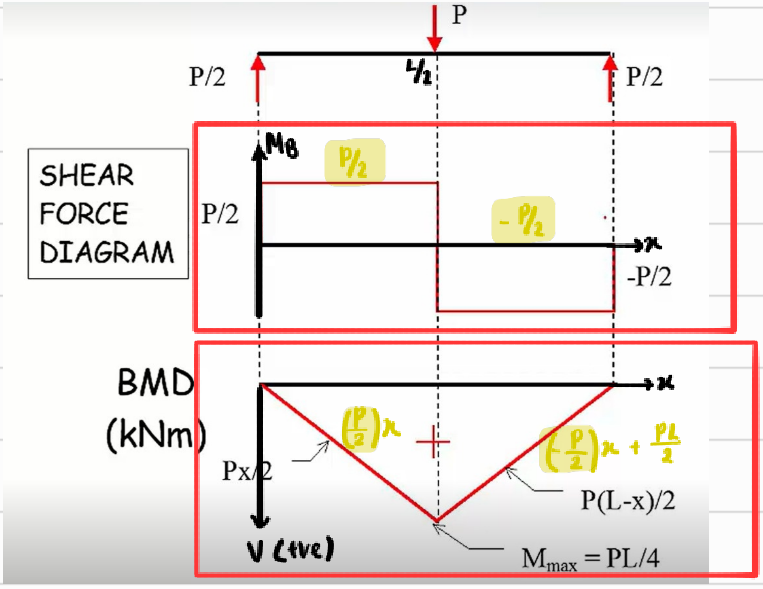
At $x = L/2$

∴ This is M_{max} in a simply supported beam of length L, carrying UDL, w



BMD shortcut

Now do notice the below

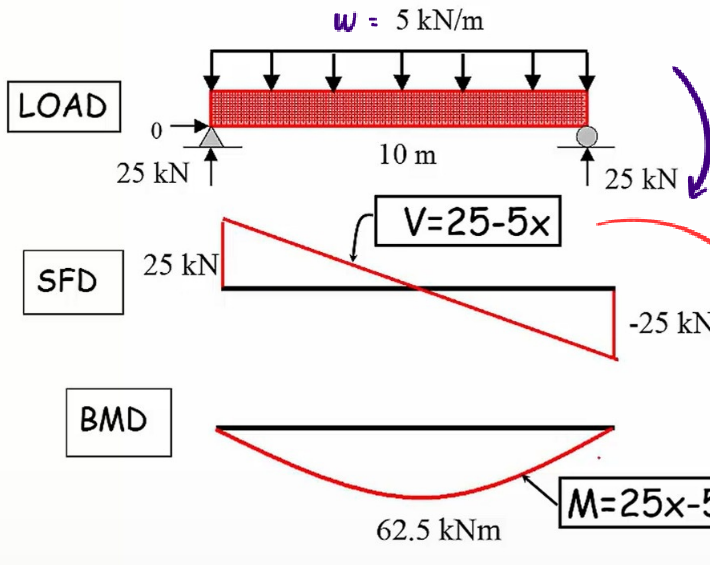


V is the gradient of M

$\boxed{V = \frac{dM}{dx}}$

*Why is the gradient of M -ve when V is +ve? Why when V becomes more negative, M becomes more positive?

∴ Rmb, our graph is plotted downwards. If our graph were plotted normally [upwards], it will be more clear



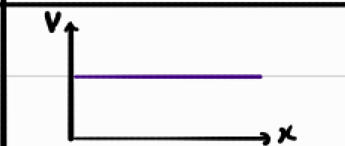
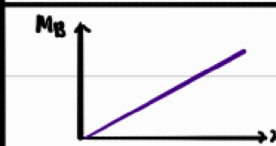
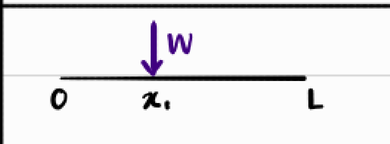
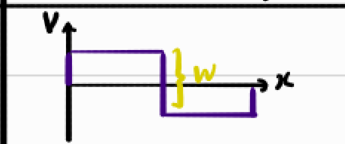
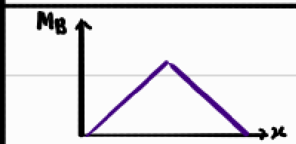
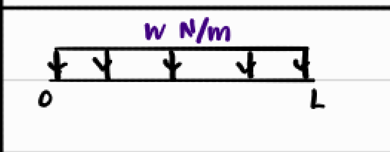
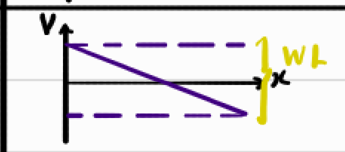
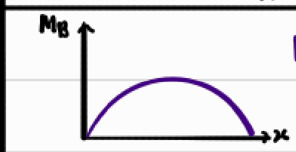
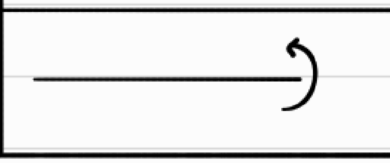
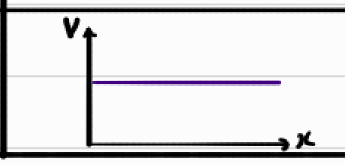
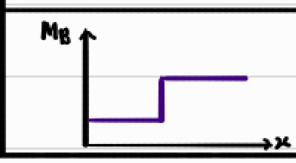
$w = \frac{dV}{dx}$

$M = \int V\,dx$

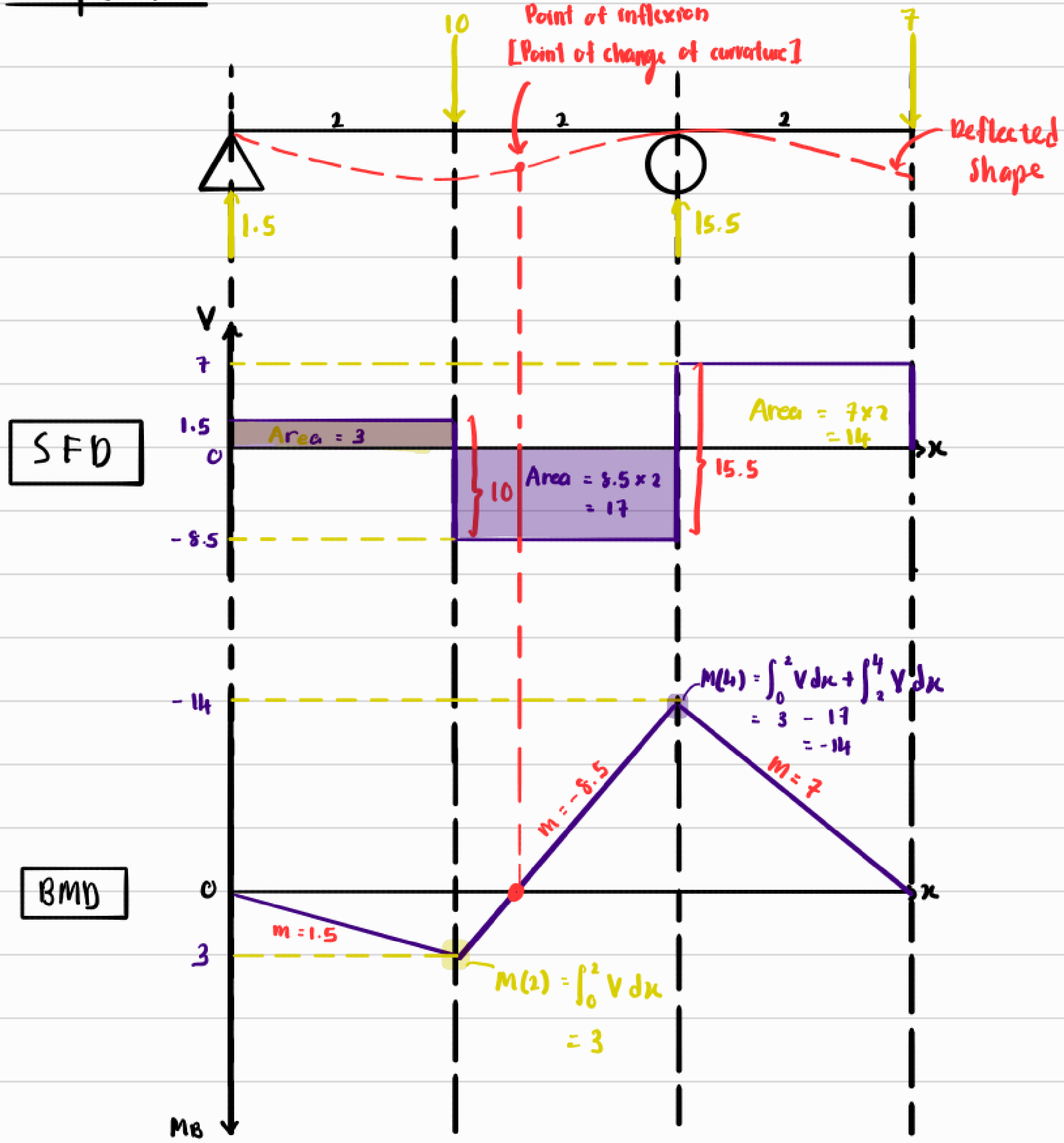
$\boxed{V = \frac{dM}{dx}}$

∴ +ve V ⟹ Downwards slope BMD

-ve V ⟹ Upwards slope BMD

Type of load	Diagram of load	SFD	BMD [If plotted upwards]
No Load			 Linear Line
Point Load			 Kink
Uniformly distributed Load			 Parabola
Applied Moment			 Step

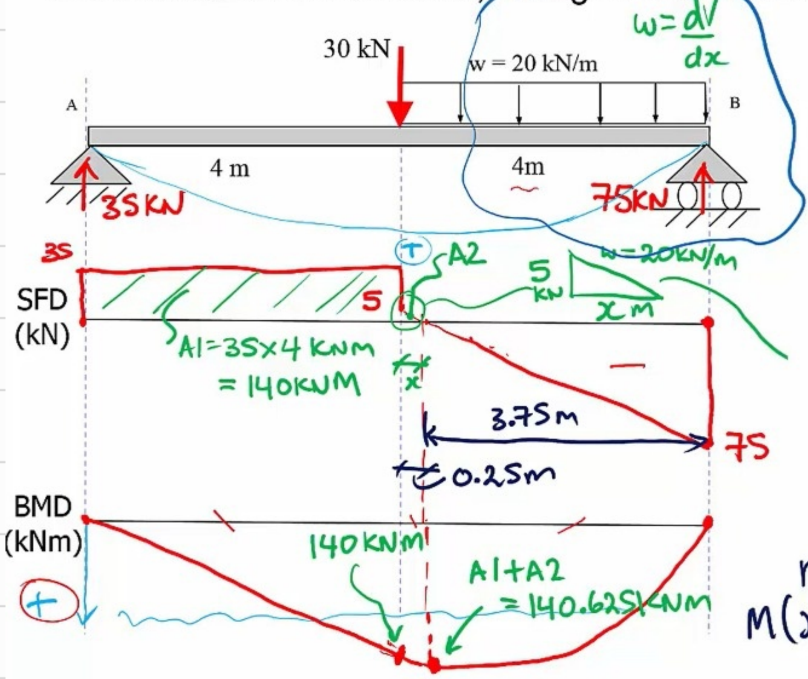
Example 1:

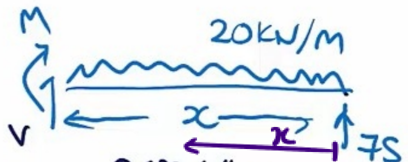


10
Point of inflexion
[Point of change of curvature]
7
2
2
2
Deflected shape
1.5
15.5
V
7
1.5
0
-8.5
SFD
Area = 3
Area = 8.5 × 2
= 17
10
15.5
Area = 7×2
= 14
x
- 14
$M(4) = \int_0^2 V dx + \int_2^4 V dx$
= 3 - 17
= -14
m = -8.5
m = 7
BMD
0
x
m = 1.5
3
$M(2) = \int_0^2 V dx$
= 3
MB

Example 2:

Draw the SFD and the BMD, noting all local maximum and minima values.





$0 < x < 4\,m$

$\Sigma F_y = 0$

$V - 20x + 75 = 0$

$V = 20x - 75$

$V = 0 = 20x - 75$

$x = \frac{75}{20} = 3.75\,m$

$\Sigma M @ cut = 0$

$-M - 20x \cdot \frac{x}{2} + 75x = 0$

$M = 75x - 10x^2$

$M(x = 3.75) = 75 \times 3.75 - 10(3.75)^2$

$= 140.625$